Atom Loss Tolerant QEC on Neutral Atom Arrays

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Abstract

Neutral atom arrays have emerged as a leading platform for scalable fault-tolerant quantum computing, but their performance is strongly constrained by atom loss. In this review, atom loss is treated as an architectural challenge for quantum error correction (QEC), particularly because loss events violate the standard fixed-qubit assumptions. It is discussed how loss can be converted into flagged erasures, enabling erasure-aware decoding and improved thresholds relative to standard Pauli-noise models. Recent hardware-efficient strategies are examined in which qubits are engineered so that dominant physical noise channels are made more correctable, and are then concatenated with codes matched to those noise characteristics. A holistic perspective is adopted in which code structure, decoder design, and noise modelling are considered jointly for neutral-atom systems. The review therefore focuses on pathways for integrating fast, reliable loss detection and recovery into scalable neutral-atom architectures while managing control overhead and implementation complexity.

1 Introduction

Since their experimental realization in the early 2000s, neutral atom arrays were largely regarded as scalable but slow, low-fidelity platforms that were better suited to quantum simulation than to universal quantum computation. In recent years, substantial progress has been achieved, including high-fidelity single-qubit gates [1], two-qubit entangling gates [2], and large-scale arrays containing hundreds of atoms [3]. As a result, neutral atom arrays have been established as strong candidates for scalable fault-tolerant quantum computing, with key advantages such as mid-circuit reconfigurability, long coherence times, and highly parallel entangling operations. For fault tolerance to be realized on this platform, high-speed operation with high-quality qubits at large scale is required. Under standard quantum error correction (QEC) approaches, however, the target error regimes are typically reached only with very large physical-qubit overheads [4, 5]. One of the key challenges in achieving this is atom loss, which can compromise the effectiveness of QEC strategies on this platform.

In neutral atom arrays, individual Rydberg atoms are trapped in optical tweezers, quantum gates are implemented with laser pulses, and readout is performed by fluorescence imaging. During these operations, atom loss is induced by collisions, spontaneous emission, and trapping imperfections [3]. In contrast to standard bit-flip and phase-flip errors, atom-loss events are often heralded: the error location is known, but the qubit state is destroyed. Consequently, conventional QEC assumptions of a fixed qubit set with hidden errors inferred only through syndrome data are violated. Although undetected loss can produce logical failure, detected loss can be converted into flagged erasures, and improved decoding performance and higher physical error thresholds have been reported relative to standard Pauli-noise settings [6, 7, 8]. The central challenge is implementing fast, reliable loss detection with low false-positive rates and overhead.

To address this challenge, hardware design now focuses on explicitly engineering qubits so their dominant noise source can be corrected more efficiently [9, 10, 11, 12, 13, 14, 8, 15, 16, 17, 18, 19]. This approach achieves significantly better error thresholds and reduces the physical-qubit overhead required for fault tolerance as hardware-efficient QEC can be tailored to the dominant noise channel, and decoders can leverage the additional information provided by heralded loss events. The main remaining bottleneck is scale: achieving useful logical error rates still requires large physical-qubit counts and substantial control overhead. A central design objective is therefore to integrate fast, reliable loss-detection and recovery directly into hardware while keeping added complexity and scaling costs manageable. This review focuses on atom-loss-tolerant QEC for neutral-atom arrays with a holistic approach considering the interplay between the code, decoder, and noise model in achieving fault tolerance. To begin the

exploration of hardware-efficient QEC tolerant to atom loss, this review will cover the background of qec and neutral-atom hardware.

1.1 QEC background

QEC refers to techniques that suppress the effective error rate of quantum computations by encoding one logical qubit into many physical qubits, then repeatedly detecting and correcting faults without directly measuring the logical state. Error correction alone is not sufficient for scalable quantum computing. Fault tolerance as described by threshold theorem requires a more holistic approach that also takes in to account the quality of the Error Model, Code, and Decoder and there exists a physical error rate below which increasing code distance suppresses logical error [20, 21, 22]. Operating below threshold is therefore the key practical criterion, and has become a central benchmark in recent experiments [23]. At a high level, the QEC cycle has three components: encoding, error detection, and decoding [24, 25, 26].

Types of Errors
Qubits are generally susceptible to a wide variety of error types, which can be broadly categorized into three classes: leakage errors, Pauli errors, and erasure errors. Leakage errors are the most harmful because they take the qubit out of the computational subspace, and are undetectable without additional hardware. Pauli errors stay within the computational subspace and can be detected and corrected by the error-correcting code. Erasure errors are initially undetectable but can be flagged by erasure checks, allowing for efficient correction.

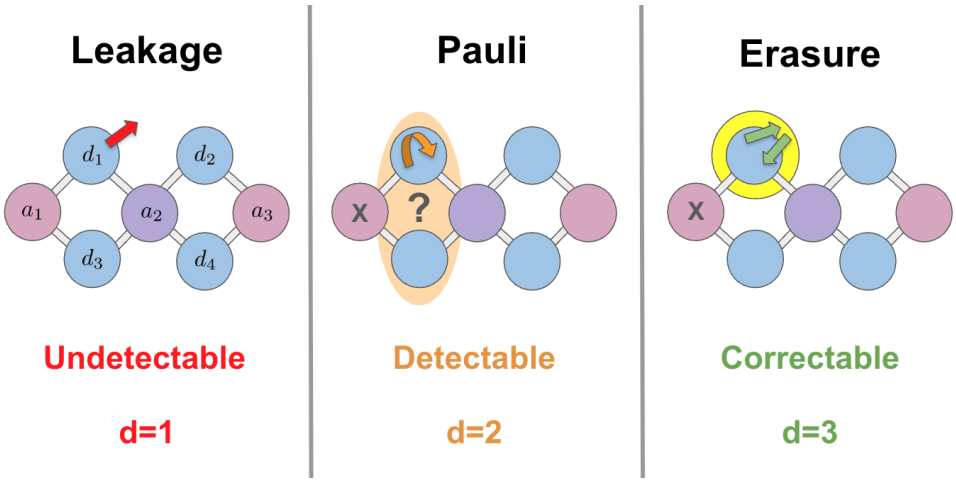

Figure 1: Comparison of leakage, Pauli, and erasure errors in a minimal surface-code layout, with data qubits d_i, ancilla qubits a_i, and effective code distance d. Leakage errors escape the computational subspace without any flag (undetectable, distance $d = 1$); Pauli errors stay within the subspace and flip stabilisers (detectable, $d = 2$); erasure errors initially escape the computational subspace, then are flagged by erasure checks and can be reset and corrected (correctable, $d = 3$).

These error types form a hierarchy of increasing detectability and correctability, with leakage errors being the most harmful and erasure errors being the most benign. Hardware-efficient QEC strategies therefore aim to engineer qubits so that the dominant noise channel is erasure errors, which can be efficiently detected and corrected, while suppressing leakage and Pauli errors as much as possible.

Encoding
Encoding maps logical states into an entangled code space so that local physical errors can be detected and corrected while preserving the encoded information. Different code families make different trade-offs between overhead, connectivity, and decoding complexity. Surface codes are currently the most mature for local 2D hardware because they rely on nearest-neighbour stabiliser checks and have high practical thresholds, but they require substantial physical-qubit overhead. High-rate qLDPC codes can encode more logical qubits per physical qubit, potentially reducing overhead, but typically demand more complex check graphs and fault-tolerant implementations.

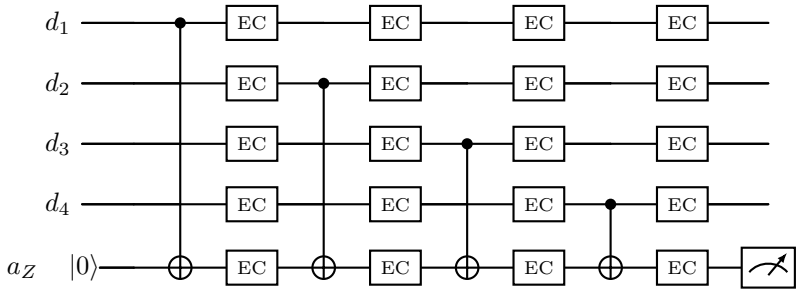

Figure 2: Syndrome extraction circuit for a $d = 2$ surface code with integrated erasure checks. The data qubits d_i and ancilla a_Z are dual-rail encoded qubits. Each erasure check flags leakage out of the computational subspace, after which the affected qubit is reset before continuing the circuit.

Decoding
Decoding is the classical inference step that maps syndrome data to a likely error pattern and a corresponding recovery operation. For topological codes, widely used decoder families include minimum-weight perfect matching and Union-Find methods, which differ in accuracy, runtime, and implementation cost. The core challenge is degeneracy: many different physical error configurations can produce the same syndrome. Effective decoders therefore combine syndrome history with a realistic noise model and any side information (for example, heralded erasure locations) to minimise logical failure probability.

1.2 QEC on Neutral atoms

Neutral-atom quantum processors trap individually addressable atoms in optical tweezers or optical lattices and encode qubits in long-lived internal states. Large, defect-reduced 2D/3D registers are built by loading and rearranging atoms with movable tweezers, and entangling gates are implemented by transiently exciting selected atoms to Rydberg levels ($n > 10$), where strong dipole-dipole interactions enable fast conditional dynamics. The main advantages follow directly from this architecture: natural atomic uniformity minimizes device-to-device variability, optical trapping supports high-density scaling and reconfigurable connectivity via qubit shuttling, long-lived encodings provide extended coherence, and laser-based control in vacuum can support mid-circuit measurement/reset without requiring the heavy cryogenic infrastructure typical of superconducting platforms.

1.2.1 Neutral atom error profile

1.2.2 Encoding on neutral atoms

To see how an erasure error-correction protocol operates in practice, consider the surface-code syndrome extraction circuit in figure 2. The four data qubits d_i form a surface code, and the ancilla a_Z measures a Z-type stabiliser that flags Z errors on the data qubits. Both data and ancilla are dual-rail encoded qubits, so erasure checks can be applied throughout the circuit.

2 Atom loss error model

3 Atom loss QEC codes

Existing qubit loss–aware code strategies for neutral-atom arrays now span four main approaches: (i) surface codes augmented with dedicated loss detection and loss-aware decoding, (ii) architectural schemes that relocate logical qubits away from accumulated erasures, (iii) fully experimental surface-code implementations that exploit loss information via “superchecks” and machine-learning decoders, and (iv) dual-rail encodings that natively map qubits onto an erasure model. Together, these works show that treating atom loss as an erasure with known location can significantly raise effective thresholds and improve logical performance, with dual-rail encodings providing a particularly clean mapping from physical

loss processes to erasure errors at the code level. In neutral-atom platforms, dual-rail approaches can be realized either via two-atom encodings or via internal-level encodings that ensure loss or leakage produces a detectable vacancy, making them highly compatible with erasure-oriented decoders.

Surface-code schemes with loss detection units (LDUs) exemplify the first class: LDUs identify lost atoms each cycle, feed explicit erasure locations to an adaptive decoder, and achieve orders-of-magnitude reductions in logical error probabilities relative to loss-agnostic decoding, with threshold lines that trade off tolerated loss versus depolarizing noise. These schemes scale naturally in code distance, require only local detection plus more sophisticated decoding, and become even more accurate when combined with dual-rail encodings or erasure-conversion techniques that convert leakage and other faults into flagged erasures, thereby aligning the physical noise more closely with the assumed erasure model.

Erasure-recovery architectures such as k-shift protocols address long-time operation in dense arrays, where erasures accumulate and naive refilling leads to vanishing effective thresholds. They periodically deform the surface code and transport atoms so that the logical patch is shifted away from damaged regions into freshly prepared ones, restoring practical logical error suppression at finite distances that are realistic for near-term devices but at the cost of code-deformation overhead and large-scale rearrangement. Dual-rail encodings can be integrated here by moving logical dual-rail pairs rather than single sites, using one rail as a sacrificial channel for loss, which stabilizes the effective erasure pattern seen by the decoder while increasing physical-qubit overhead.

Large-scale neutral-atom experiments implement surface codes with explicit loss-aware decoding by forming "superchecks" (products of stabilizers around lost atoms) and training most-likely-error or machine-learning decoders on loss patterns, thereby recovering syndrome information and reaching below-threshold logical performance in multi-round QEC experiments. These demonstrations validate that loss-aware strategies are experimentally viable at the few-hundred–atom scale and provide a natural setting for dual-rail encodings, where decoders can directly exploit structured correlations between rail occupations, loss events, and stabilizer outcomes [27].

Overall, LDUs and superchecks with loss-aware decoding appear most accurate and scalable as code-level treatments of atom loss, while dual-rail encodings and erasure-conversion schemes are most effective at cleaning up the physical noise so that it matches an erasure model. Erasure-recovery via k-shift–type protocols becomes important when loss is strong and densely packed arrays limit local refilling, and it can be combined with dual-rail encoding to make the motion of logical information more robust. The emerging consensus is that the most accurate, scalable neutral-atom architecture for atom loss will combine dual-rail or erasure-converted qubits, real-time loss detection (LDU-like mechanisms), superchecks plus loss-aware decoding, and periodic patch relocation when erasure density becomes too high.

4 Atom loss Decoding

Loss-aware decoding strategies have emerged as a crucial ingredient for achieving fault tolerance in architectures where qubit loss or erasure is a dominant error channel, including neutral-atom arrays. These approaches exploit additional information provided either by active loss detection units (LDUs), by encoding (dual-rail), or by code-deformation tools such as superchecks and k-shift protocols, to convert loss into a structured erasure pattern that can be handled more efficiently than generic Pauli noise. At a high level, all of these strategies trade hardware or control overhead for a substantial reduction in logical error rates at realistic loss probabilities, but they differ in how aggressively they use loss information and in how forgiving they are to imperfect detection.

Strategies based on loss detection units integrate additional measurements into the surface-code cycle to flag lost atoms before or during stabilizer extraction. In the atom-loss–resilient surface-code construction for neutral atoms, standard LDUs and teleportation-based LDUs are compared via circuit-level simulations under combined atom loss and depolarizing noise. Both variants feed explicit loss locations into a loss-aware decoder, which effectively works on an erasure channel overlaying a Pauli channel. Simulations show that including LDUs can improve logical error probabilities by orders of magnitude at fixed physical error rates, and that an atom-loss threshold on the order of a few percent is achievable when

depolarizing noise is sufficiently low. Teleportation-based LDUs tend to be more robust to detection-induced back-action but require additional ancillae and gates, whereas standard LDUs are simpler but somewhat more sensitive to measurement errors.

Dual-rail encoding takes a different route by embedding loss detection directly in the logical qubit definition. In cavity and superconducting implementations, the logical states are encoded in single-excitation subspaces of two modes, so that dominant photon-loss events manifest as erasures that can be detected via syndrome-like checks or post-selection. In dual-rail cavity qubits, error-detection protocols convert most physical amplitude damping into flagged erasures, leaving a background of Pauli-like errors that is orders of magnitude smaller than the original noise. Experiments with dual-rail encoded multimode transmons demonstrate logical bit-flip and phase-flip error rates more than an order of magnitude lower than the physical ones, with stability over hours-long runs; this confirms that hardware-level erasure detection can deliver a highly favorable hierarchy of error rates for downstream QEC decoders. While these works do not implement large-distance codes, they indicate that when loss is the dominant channel, dual-rail encoding can supply exceptionally clean erasure information to any standard erasure-aware decoder.

In neutral-atom surface-code experiments, superchecks and related loss-aware decoding schemes have been used to compensate for atom loss during stabilizer measurements. A recent fault-tolerant neutral-atom architecture demonstrates the use of superchecks, constructed by multiplying stabilizers surrounding a lost atom, to recover parity information that would otherwise be missing. The loss pattern is then incorporated into both maximum-likelihood and machine-learning decoders trained on simulated and experimental data. The authors compare different strategies-treating loss as $|0\rangle\,|0\rangle$, explicit loss detection without superchecks, supercheck construction, and post-selection on fully populated detectors—and show that superchecks combined with loss-aware decoding significantly suppress the detected error probability relative to loss-ignorant schemes. This confirms earlier theoretical predictions that exploiting erasure information in the decoder can strongly enhance surface-code performance in the presence of atom loss.

The k-shift erasure-recovery protocol for neutral-atom surface codes pushes loss awareness further by combining code deformation with physical rearrangement of atoms. In this scheme, erasures (lost atoms) are corrected not simply by local superchecks, but by repeatedly transferring the logical qubit from an imperfect array with accumulated erasures to a pristine one, while correcting erasures in the emptied region using tweezers and atom transport. The decoder operates on dynamically deformed codes where stabilizers are redefined to avoid erased sites; numerical simulations show that k-shift erasure recovery restores the threshold behavior of the surface code in regimes where accumulated erasures would otherwise destroy its error-correcting capability. Compared with simply repeating syndrome measurements on a static array, incorporating k-shift yields a substantially reduced logical failure rate at fixed physical loss and depolarizing rates.

Comparing these approaches, the most effective strategies are those that (i) achieve high-fidelity, near-real-time identification of loss events, and (ii) feed this information into a decoder explicitly designed for erasures rather than generic Pauli noise. LDUs on neutral-atom surface codes achieve this with relatively modest overhead and already reach atom-loss thresholds of a few percent, especially when combined with teleportation-based detection that limits back-action. Superchecks are particularly powerful in experimental settings because they allow standard decoders (maximum-likelihood or machine-learning based) to operate on effectively complete syndrome information even in the presence of loss, leading to clear reductions in detected error rates. The k-shift protocol appears most advantageous at higher steady-state loss rates, where static-patch strategies fail, but its performance gains come with significant control complexity due to repeated code deformations and atom transport. Dual-rail encoding provides arguably the clearest erasure channel and the lowest logical error per physical loss event, but it shifts the overhead into hardware resources (extra modes or qubits per logical qubit) and thus is most natural for platforms where bosonic or multimode structures are readily available. Overall, for qubit-loss–aware codes in neutral-atom arrays, hybrid schemes that combine local LDUs and superchecks with erasure-tailored decoders look most practical near-term, while k-shift protocols offer a promising route to maintain fault tolerance in regimes of sustained, high atom loss.

5 Conclusion

5.1 Open questions and avenues for progress

We have discussed several open questions which could lead to interesting further explorations in the near term. These include the possibility of modifying hardware designs to further optimise the advantage to be gained from engineering erasure noise, discussed in section 1.2.1, and the unknowns regarding which codes stand to gain most significantly from enabling erasure-based error correction.

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